

Aadesh Umesh Gund

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PROFESSIONAL SUMMARY

Computer Science student at VIT-AP with hands-on experience in Machine Learning, Artificial Intelligence, Computer Vision, and Full-Stack Development. Proficient in Python, PyTorch, Scikit-learn, NumPy, Pandas, and modern software development tools. Built projects involving Physics-Informed Neural Networks (PINNs), traffic optimization using computer vision, and predictive AI systems. Passionate about applying machine learning techniques to solve real-world problems and continuously exploring advances in AI research.

TECHNICAL SKILLS

Data Science & ML: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, YOLOv8, Physics-Informed Neural Networks (PINN), Jupyter

Languages: Python, Java, SQL, JavaScript (ES6+)

Web Development: React, Node.js, Express.js, FastAPI, Spring Boot, HTML5, CSS3

Databases: MySQL, MongoDB

Tools & Platforms: SUMO, TraCI, Git, GitHub, VS Code, IntelliJ IDEA, Anaconda, Vercel

Core Concepts: Advanced Data Structures & Algorithms, Object-Oriented Programming (OOP)

PROJECTS

AeroFind

Python, PyTorch, DeepXDE, NumPy, Pandas, Scikit-learn

- Developed a predictive machine learning model to track and locate lost satellites by analyzing orbital history, velocity, and trajectory data.
- Implemented Physics-Informed Neural Networks (PINNs) to integrate physical laws into the model loss function, enhancing localization accuracy for complex, non-linear orbital dynamics.
- Engineered an automated data pipeline using Pandas and NumPy to preprocess large-scale satellite telemetry data, predicting future position coordinates with high precision.

FlowSync

YOLOv8, SUMO, TraCI, FastAPI, React, Python

- Designed and built an AI-driven traffic controller system utilizing robust software system design principles to optimize real-time vehicle flow and mitigate urban congestion.
- Integrated YOLOv8 computer vision models for accurate real-time object detection and vehicle density monitoring from simulated camera feeds.
- Utilized TraCI API to bridge real-world traffic logic with the SUMO simulation environment, dynamically altering traffic signal timings.
- Developed a high-performance FastAPI backend and a responsive React dashboard to visualize real-time traffic metrics and system efficiency.

ProgramEnergy

MongoDB, Express.js, React, Node.js (MERN Stack)

- Created a full-stack, comprehensive developer productivity suite featuring modular project management tools and custom backend system design.
- Architected "DevTrack", a module utilizing custom data tracking and secure JWT authentication to protect and monitor individual coding progress, time blocks, and milestones.
- Developed "CodeBattle", an interactive competitive coding environment supporting real-time socket-based challenge synchronization.
- Designed a scalable, decoupled MERN architecture ensuring seamless data synchronization and low-latency database queries.

EDUCATION

Vellore Institute of Technology (VIT) - Andhra Pradesh

Amaravati, AP

Bachelor of Technology in Computer Science and Engineering | Sep. 2024 – Present

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning Fundamentals, Artificial Intelligence Conceptual Fundamentals.

CERTIFICATIONS & ACHIEVEMENTS

- Goldman Sachs Hackathon:** Cleared Phase | June 2026
- Competitive Programming:** LeetCode Rating: 1759 (Top 10% Global Contest Participant)
- Oracle Java Foundations Associate:** Professional Certification | May 2026
- Introduction to Large Language Models:** Google Cloud Certification | June 2026
- Introduction to NLP Concepts:** Microsoft Certification | June 2026
- Introduction to Generative AI and Agents:** Microsoft Certification | May 2026
- HackerRank Problem Solving:** Skills Verification | Aug. 2025

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

GeeksforGeeks Student Chapter, VIT-AP

Aug. 2025 – Present

Tech Geek Team Core Member

- Organized and facilitated technical hackathons and coding events, boosting student developer engagement across campus.
- Collaborated with a cross-functional team to deliver hands-on workshops on full-stack development, Java, and programming tools.
- Cultivated a robust peer-learning culture by mentoring junior students in foundational DSA and modern web development stacks.